

Chapter 4 Structure of Ceramic Materials

Surfaces, Interfaces, Grain/Phase Boundaries

Surface

Why we care surface for ceramic materials ?

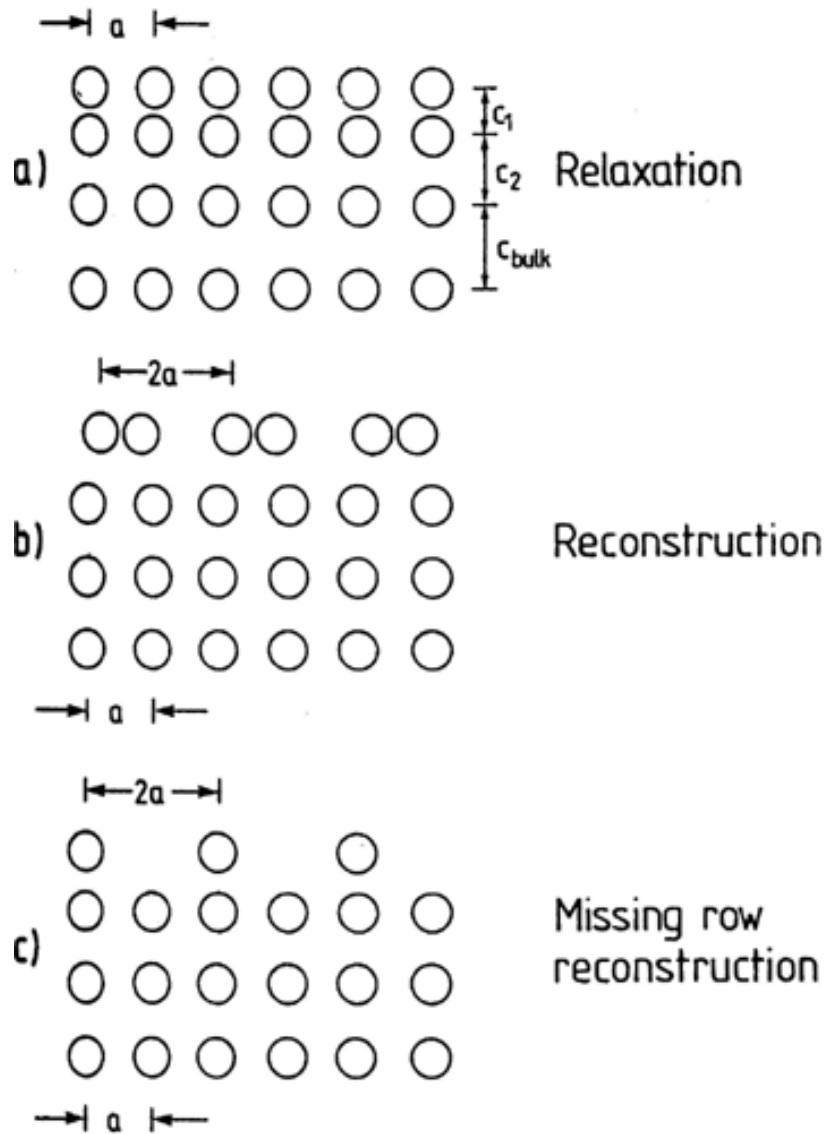
At some stage of processing, we often use powders, Surface area/Volume is larger for powders than for bulk materials

Definition: A surface is just the interface between a solid (or liquid) and a gas or vacuum. In general, the surface of a material, or any interface between materials, is a region of excess energy relative to the bulk or matrix.

Features of surfaces

1. Pressure difference;
2. The structure of surfaces can relax from the bulk terminated configuration;
3. Surface can be wetted with a thin layer of impurity or a second phase;
4. The ionicity or covalency and crystallography can affect surfaces

Relaxation, Reconstruction, Defects



Due to lack of the neighboring atoms for the surface atoms, there are characteristic rearrangement of atoms on the surface. The change of surface structure of semiconductor is more dramatic than metal.

Three kinds of this rearrangement can be seen in the shown figure: relaxation, reconstruction and missing row reconstruction.

Surface Energy

Surface energy quantifies the disruption of intermolecular bonds that occur when a surface is created.

Gibbs free energy:

$$dG = -SdT + VdP + \gamma dA + \sum \mu_i dn_i$$

Surface energy:

$$\gamma = \left(\frac{\partial G}{\partial A} \right)_{P,T,n_i}$$

Assume the binding energy of an atom is the result of bonds to its nearest neighbors

$$\varepsilon = \frac{\Delta H_s}{0.5ZN_A}$$

ΔH_s is the molar enthalpy of sublimation (the energy to break all bonds)

$$w = \frac{x}{2}\varepsilon = \frac{x\Delta H_s}{ZN_A}$$

The surface energy, γ ,

$$\gamma = \frac{x\Delta H_s}{ZN_A} \left(\frac{N}{A} \right)$$

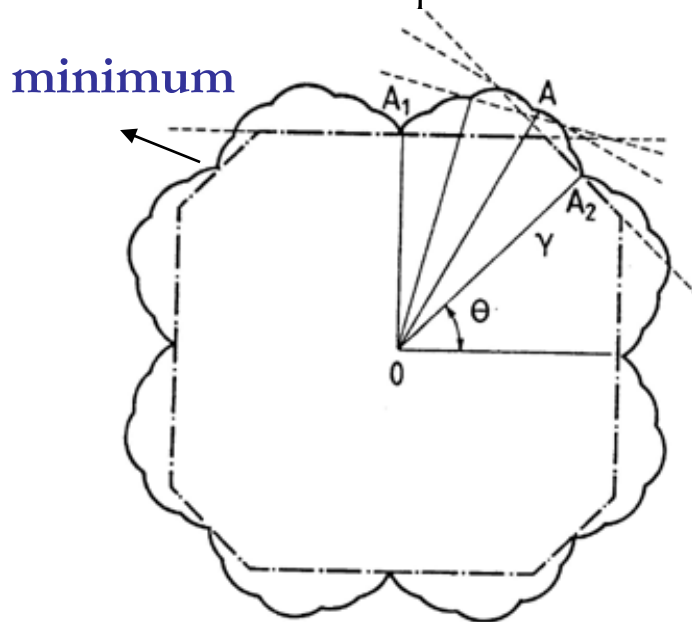
Wulff Construction

Liquids and amorphous solids have isotropic γ . Therefore they have spherical shape.

For crystalline material, γ depends on the orientation (hkl) of the surface, while the broken bonds and charge compensation per unit area may be different. $\gamma(\mathbf{n})$ is then the function of orientation of the surface, $\mathbf{n}$. It must have:

$$\int \gamma(\mathbf{n}) dA = \text{minimum}$$

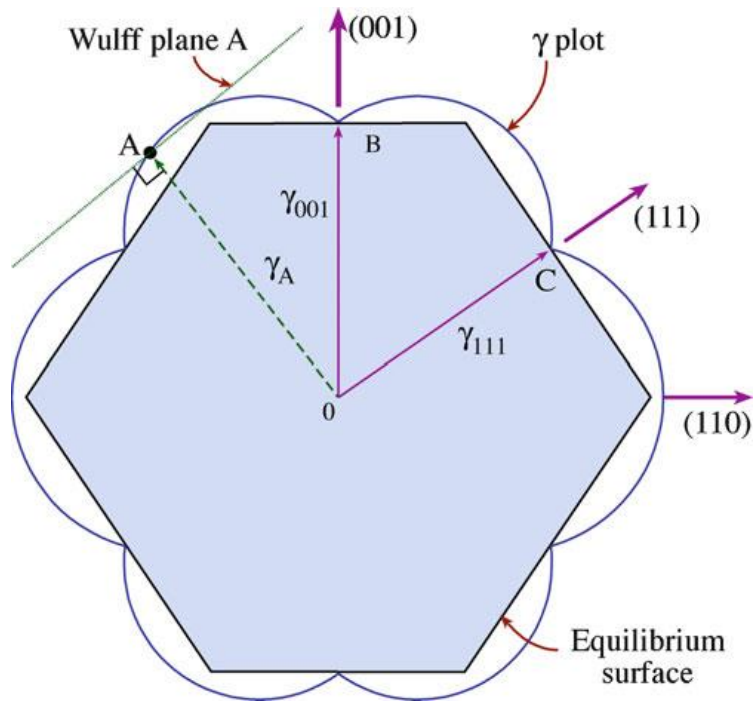
When we plot $\gamma(\mathbf{n})$ as function of n (Wulff plot), theoretically one can determine the equilibrium shape of a solid.



$\gamma(\text{hkl})$ as function of θ (the angle between the normal directions of the surface to the {hkl} planes). The Wulff plot gives the shape of the solid (dashed dotted) as the inner envelope of the Wulff planes (broken lines).

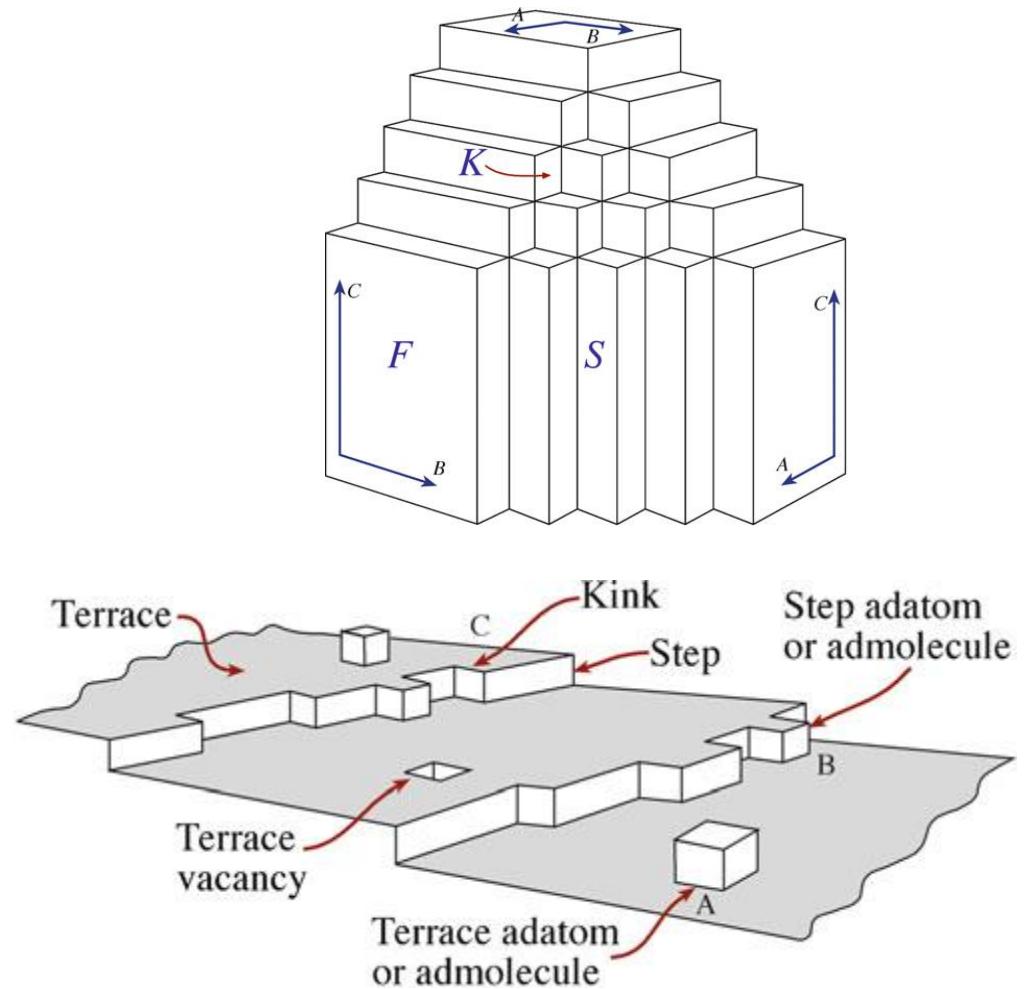
Epitaxy and film growth

Effect of Orientation



Wulff plot

Terrace step kink model



Interfaces in Polycrystals

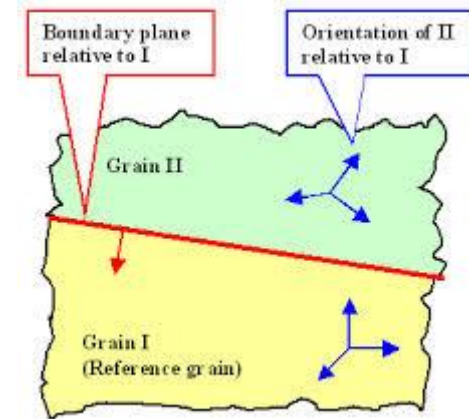
Grain boundary: the interface between two grains, or crystallites, in a polycrystalline material.

TWIST, TILT, TWIN, LOW, HIGH & 'GENERAL'

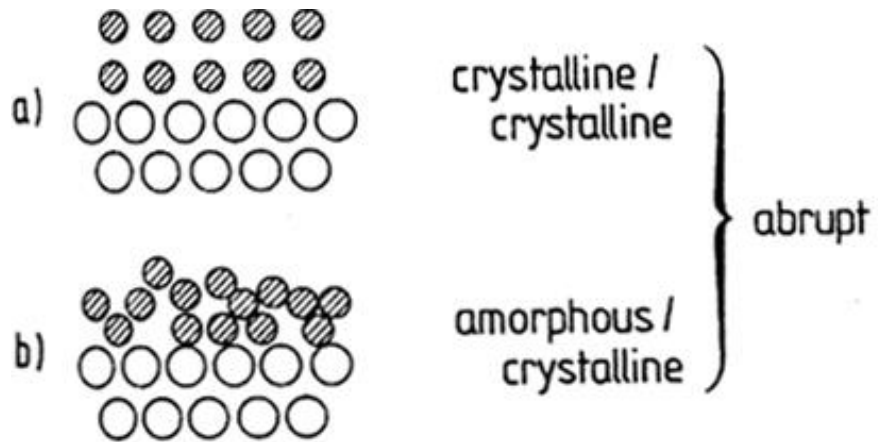
Twist	n normal to plane
Tilt	n parallel to plane
Twin boundary	Mirror across GB plane
Low-angle GB	$\theta < 10^\circ$ (Low is small!)
High-angle GB	Structured, $\theta > 10^\circ$
General GB	Not a special GB!

Features of GB: 1) The density of atoms in GBs < bulk crystals;
2) The chemistry, properties and energy differ from the bulk

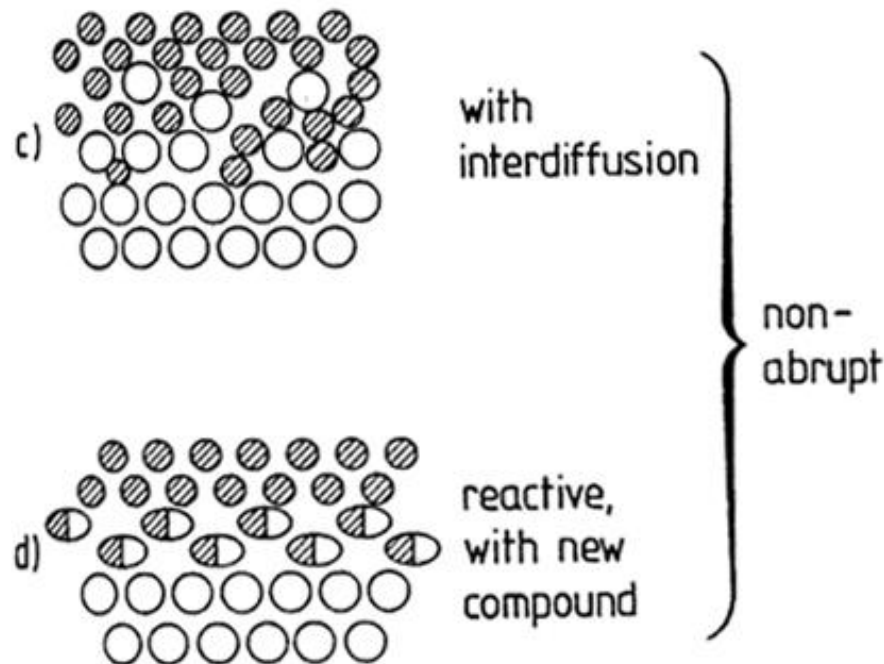
For Ceramics, 1) Localized charge or covalent bonds;
2) Local changes in density at the interfaces;
3) Impurities segregate to the GBs
4) The interface can be Structured or Amorphous;
5) The structure of GBs in Ceramics less dense than in Metals;



Structural Model of Solid/Solid Interface



The important feature of the solid/solid interface is the structure and “abruptness”. The Thermodynamics can give rough idea. Generally the interface is more complicated than the surface.



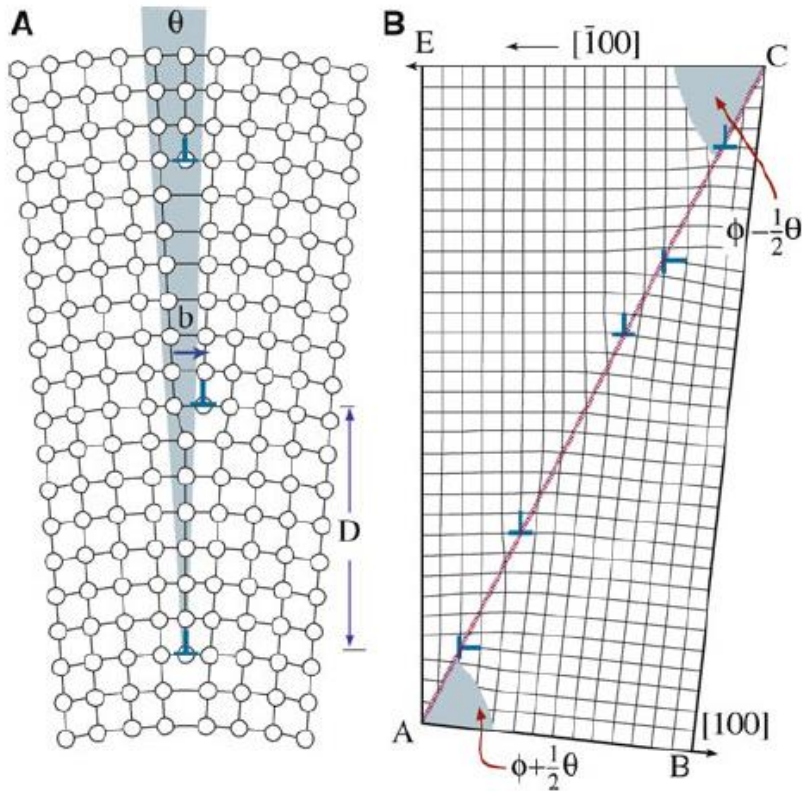
One of the important concept for the film growth is mismatch: difference between the lattice constants of the two materials, usually $(b-a)/a$ with a for substrate and b for the overlayer. The interface depends strongly on both mismatch and thickness.

Low Angle GBs

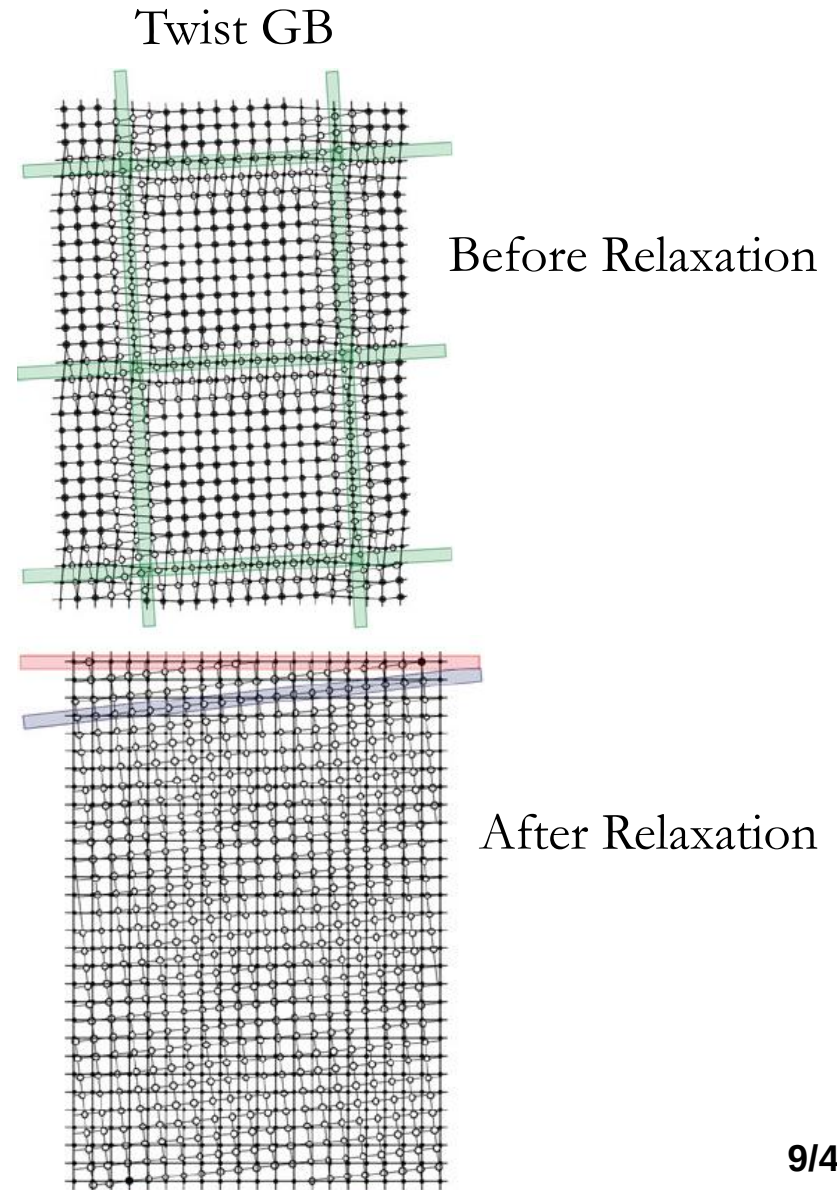
GB Energy:

$$E = E_0\theta(A - \ln\theta)$$

Low angle tilt GBs

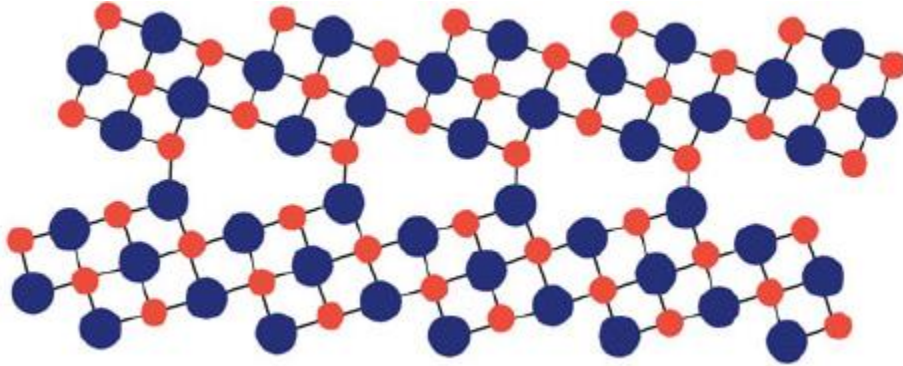


A: Symmetric; B: Asymmetric

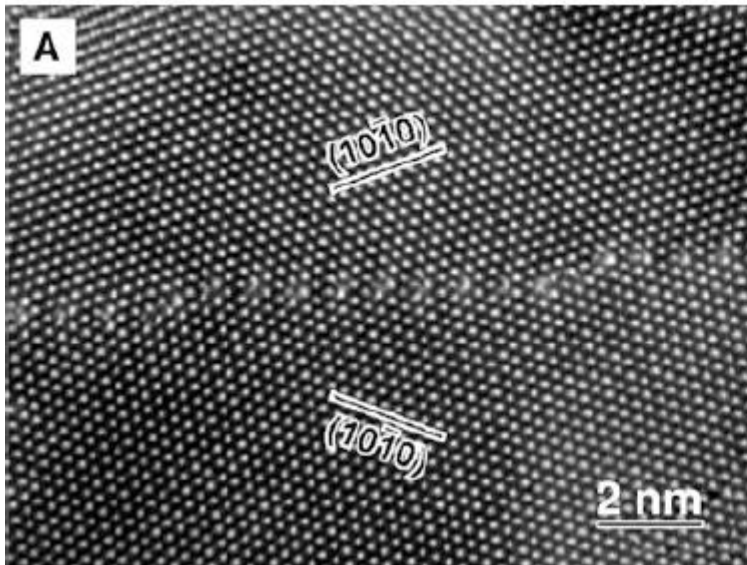
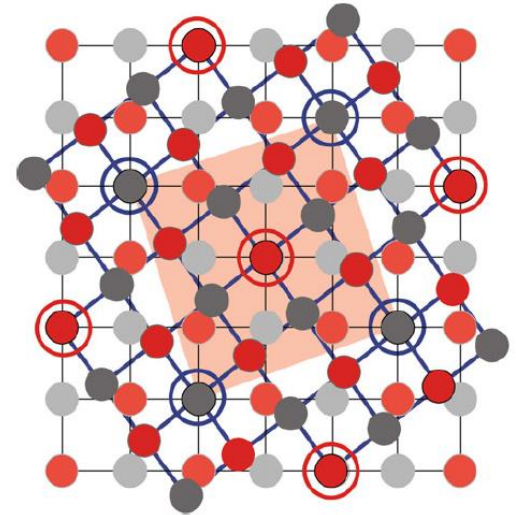


High Angle GBs

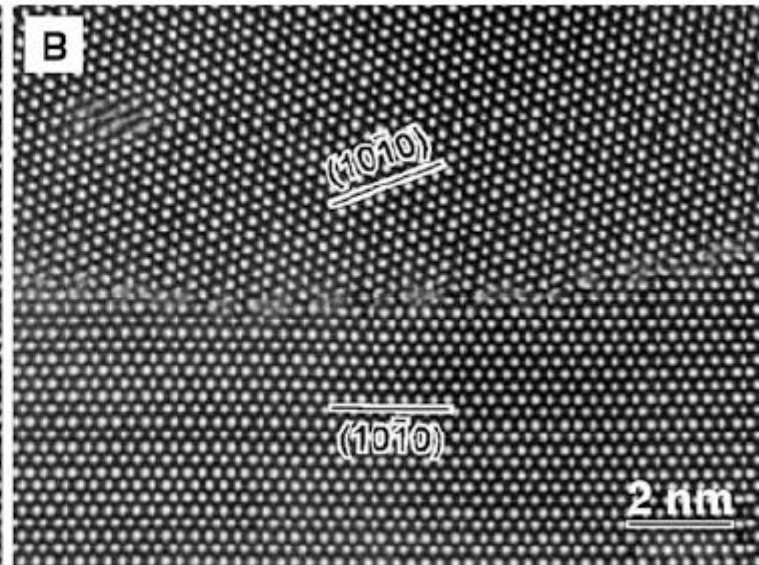
Tilt GB



Twist GB

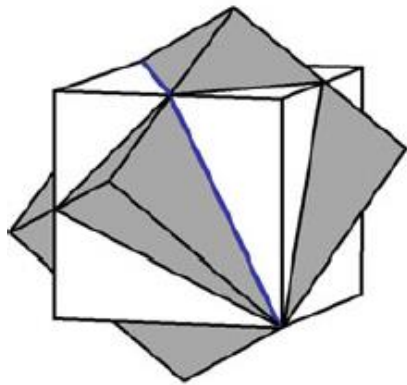


A: Near Symmetric

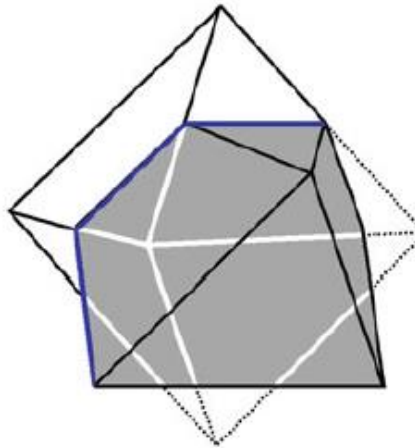


B: symmetric;

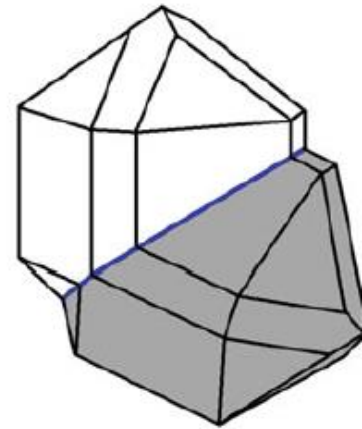
Twin Boundaries



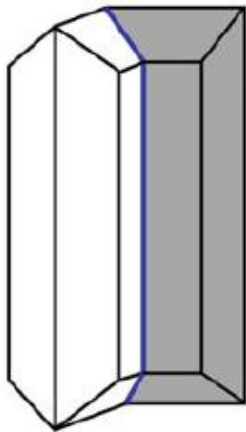
Fluorite (cubic).
Interpenetrant cubes
Rotate 60° about $\langle 111 \rangle$



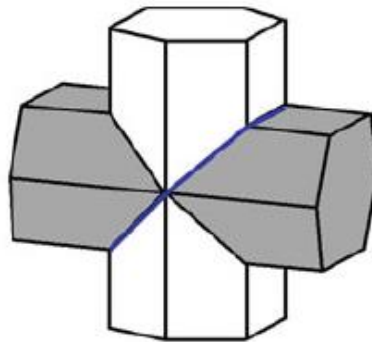
Spinel (cubic).
Contact type: twin across $\{111\}$
Can start with the octahedron shown



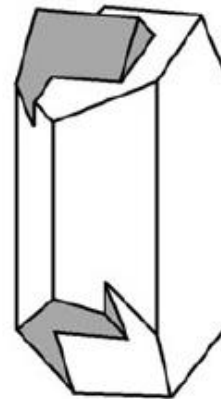
Cassiterite (tetragonal).
'Knee-twin'. Contact type
Twin plane is (011)



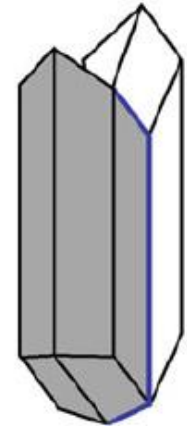
Aragonite (orthorhombic)
Contact twin
 $(\bar{1}10)$ twin plane



Staurolite (orthorhombic)
Interpenetrant
Twin-plane (032)



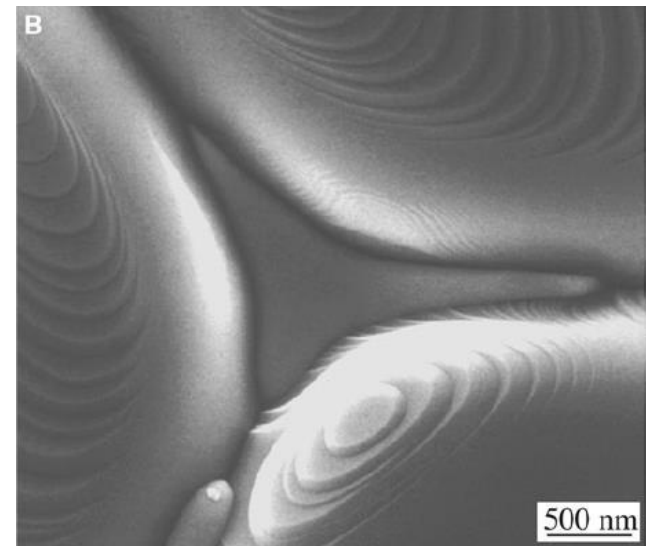
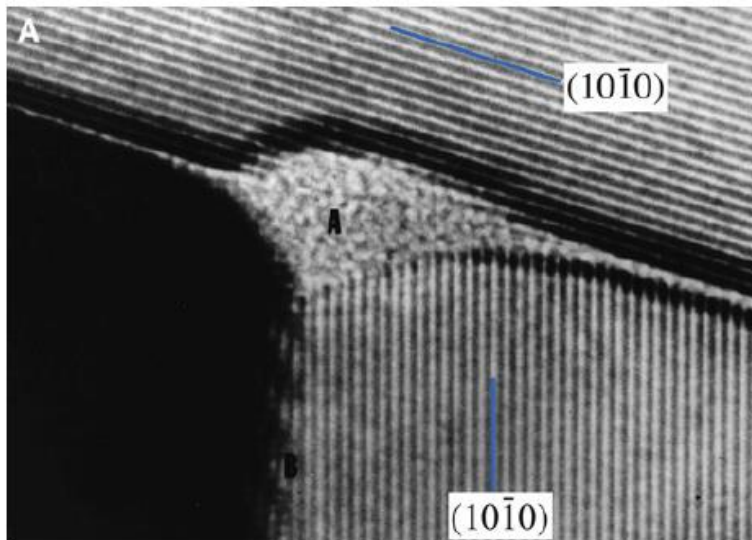
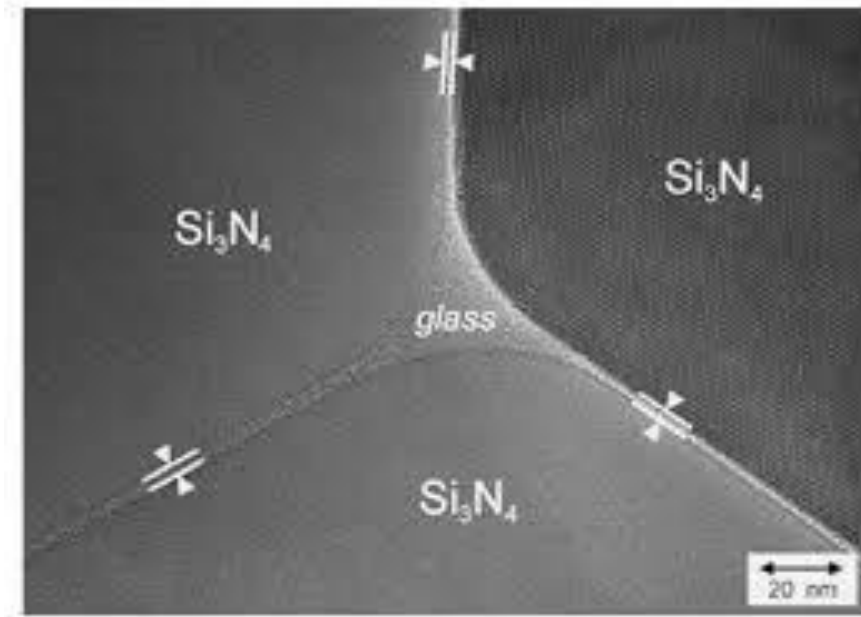
Orthoclase (monoclinic)
Twin axis is z . The Carlsbad twin
Interpenetrant across 010 .



Gypsum (monoclinic)
Swallow-tail twin
Composition plane (100)

Triple Junctions

1. One composition: three-grain junctions;
2. Two compositions: two phase boundaries plus a GB;
3. Three compositions: three phase boundaries



Phase Boundaries

PB: the interfaces between dissimilar phases. **Structurally/Chemically different.**

The Importance:

Most ceramics contain > 1 phase, thus many PBs

Phase transformations take place by the movement of PBs;

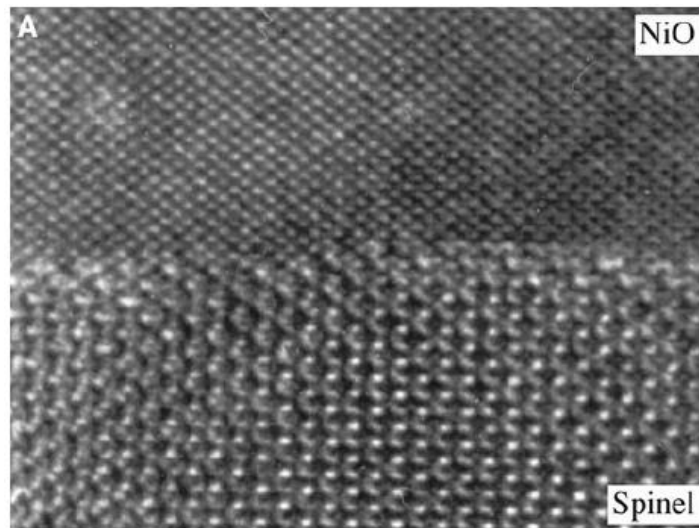
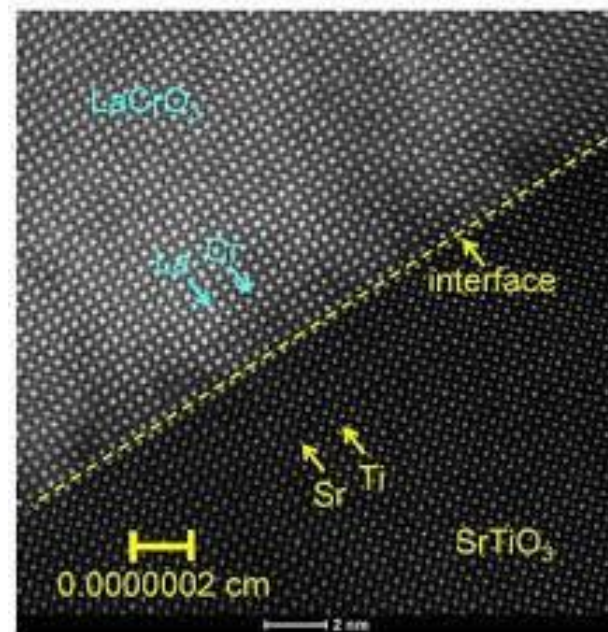
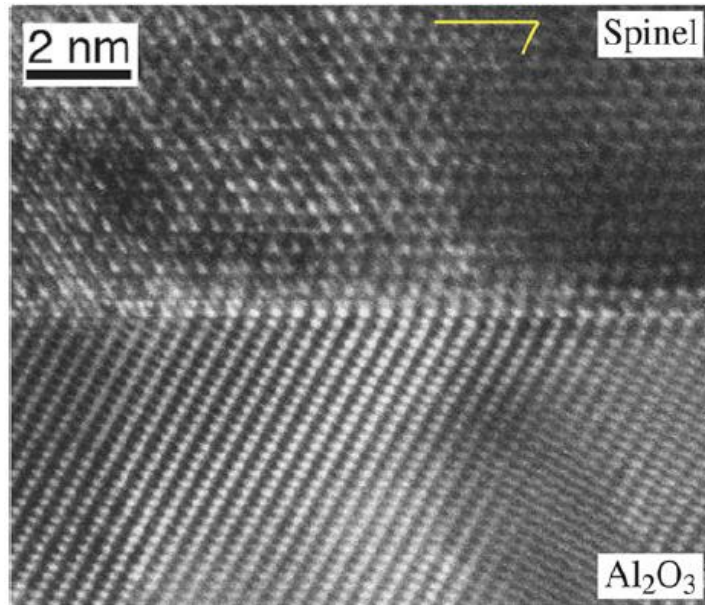
Growing one material on a different material generates a PB;

Bonding one material to another generates a PB;

<i>Bond</i>	<i>Material</i>
Brazing	Joins a ceramic to a metal
Metal-matrix composites	Makes a two-phase material
Electronic interconnects	Bonding metal to a semiconductor
Oxidation of metal	Forms oxide from metal
Catalysis	Supports catalyst particles

Formation of PB: Film growth & Processing; **Types:** Particles/Pores

Ceramics/Ceramics PBs



Z: ZrO_2
 C: CoO
 S: Co_3O_4

Nucleation and Growth of Particles

Two particles that are located within the matrix can always lower the surface energy by coalescing to form a single particle with the same volume but less surface.

- What is the driving force ? (Thermodynamics)
- How long does it take ? (Kinetics)
- What is the influence of Crystallinity ?

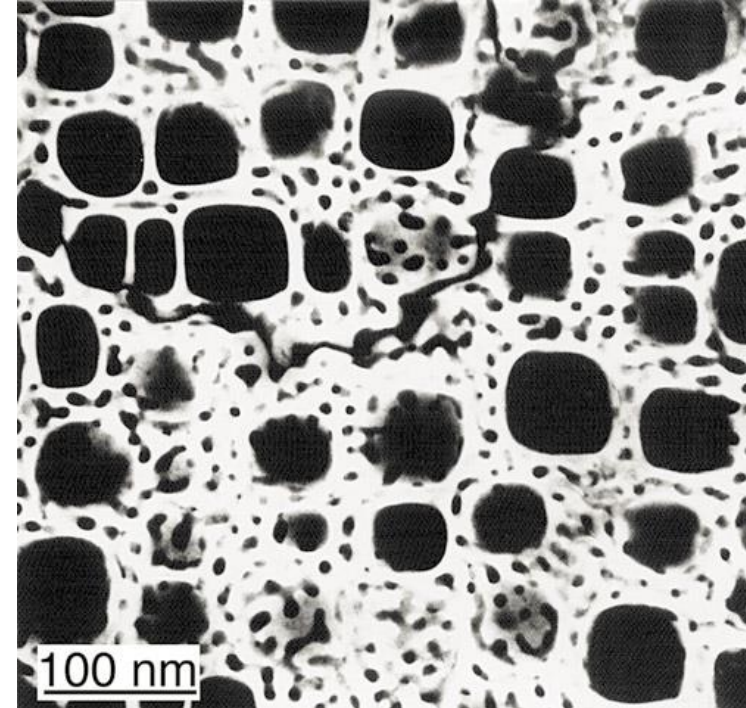
$$\Delta G_r = 4\pi r^2 \gamma_{sl} + \frac{4}{3} \pi r^3 \Delta G_v$$
$$\frac{\partial(\Delta G_r)}{\partial r} = 8\pi r \gamma_{sl} + 4\pi r^2 \Delta G_v = 0$$

The critical particle radius

$$r^* = \frac{-2\gamma_{sl}}{\Delta G_v}$$

The critical value of ΔG^* is then

$$\Delta G^* = \frac{4}{3} \pi r^{*2} \gamma_{sl} = \frac{16}{3} \pi \frac{\gamma_{sl}^3}{\Delta G_v^2}$$



Pores (Gas-Solid PB) and Measuring Porosity

Origin: Powder processing; pores can be treated as special particles.

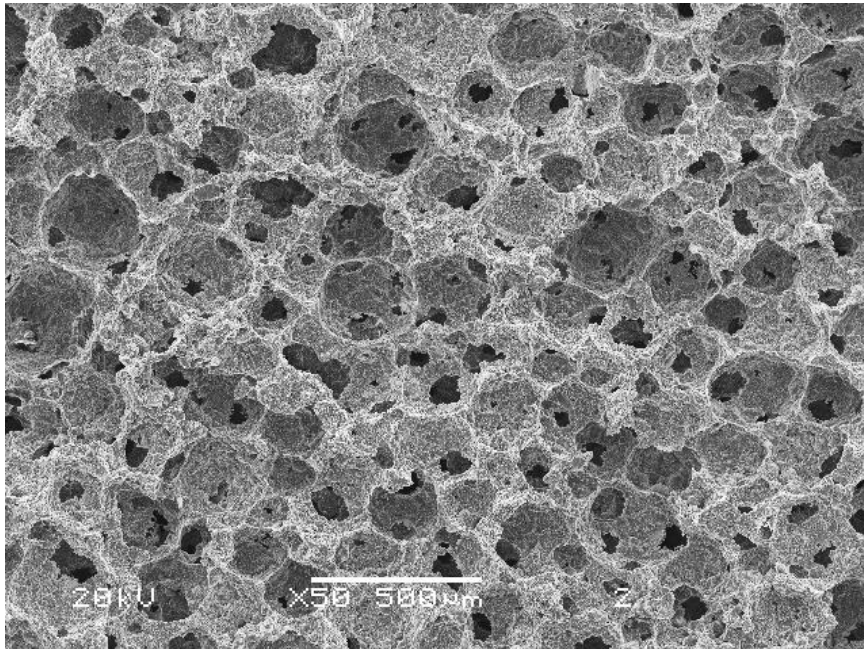
Characteristics: Curved surface, volume defects;

Parameters: pore size, the distribution of the porosity, the total surface area;

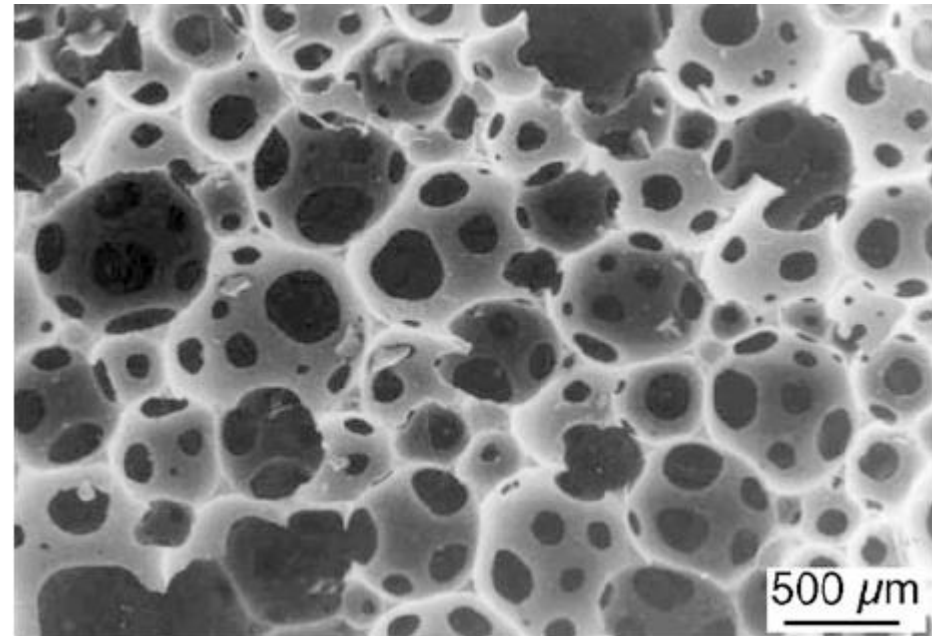
Measuring Porosity: 1) Density; 2) Mercury-Intrusion Porosimetry (压汞法) ;

$$r = \frac{2\sigma \cos\theta}{P}$$

Porous Hydroxyapatite

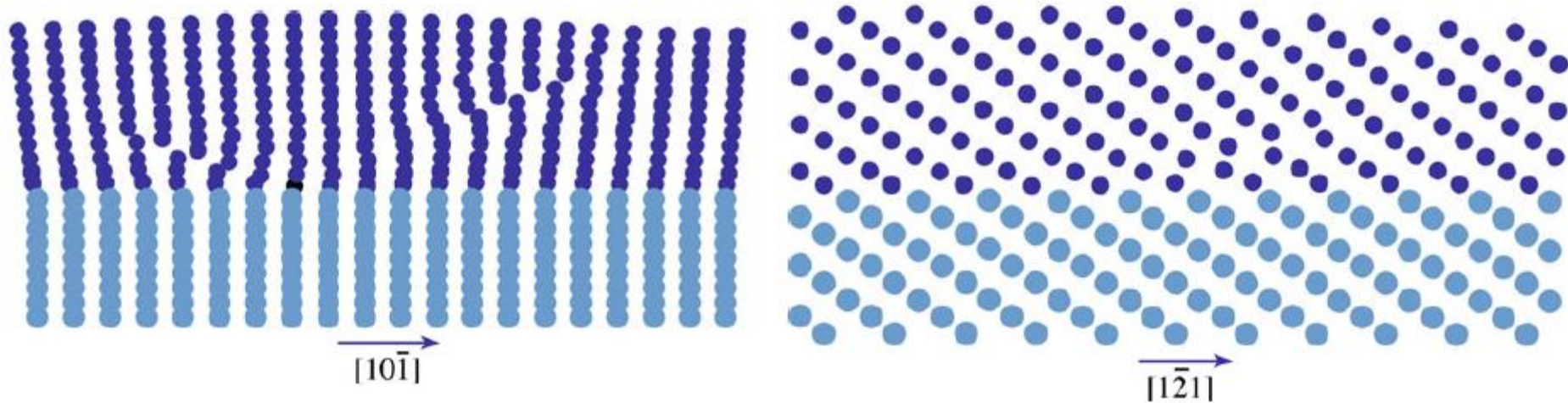


Porous Alumina



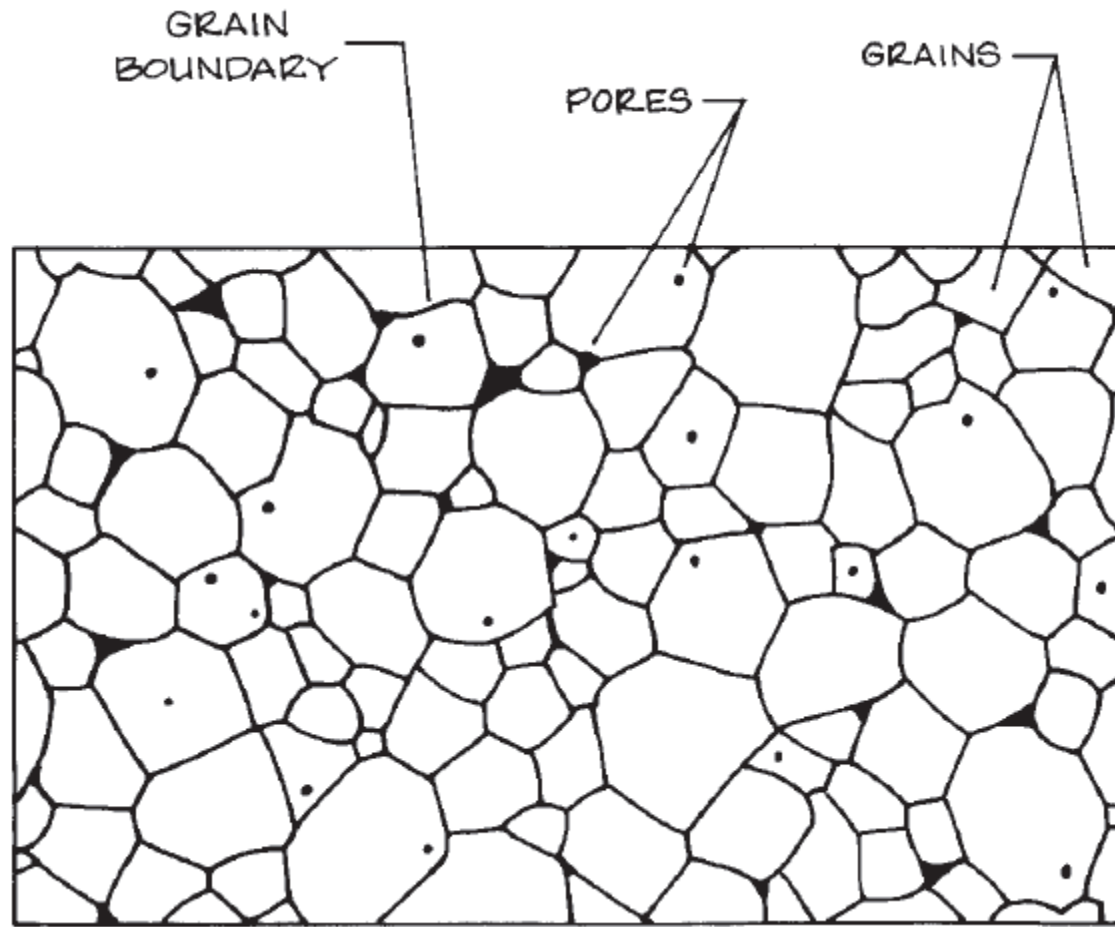
Metal/Ceramic PBs

Ceramic-reinforced Metal-Matrix Composites; when metals are oxidized; when oxides are reduced to metals; when metals grown on ceramic substances.

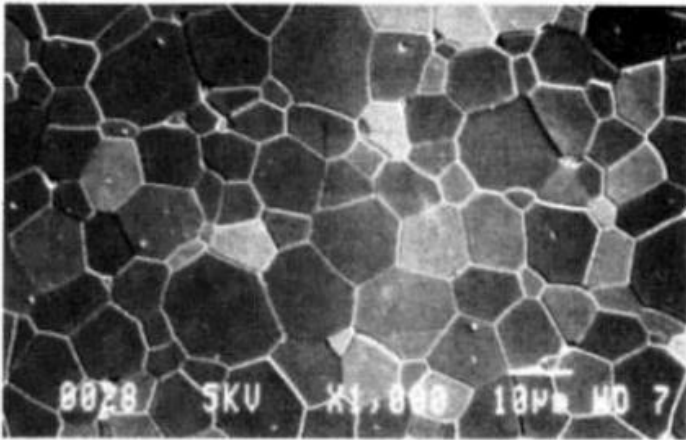


Stand-off dislocations accommodating lattice misfit between Nb and Al_2O_3 .

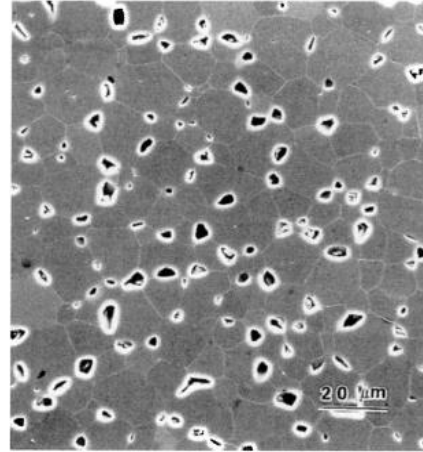
Microstructure of a typical polycrystalline ceramic



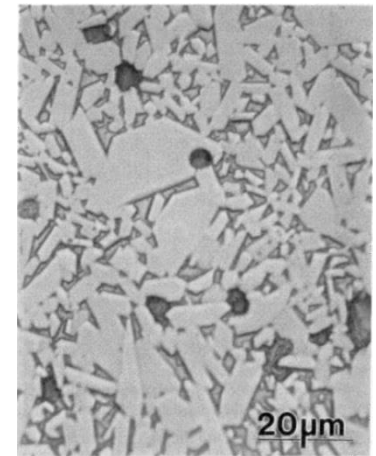
Ceramic Microstructures



(a)



(b)



(c)

- a) SSS 99% Al_2O_3 – transparent fully dense (“ideal”) ceramic: grains + grain boundaries;
- b) SSS 99% Al_2O_3 - ceramic with residual porosity: grains + GB. + pores ;
- c) LPS 96% Al_2O_3 - dense ceramic with grain boundary glass phase: grains + glassy phase ($\text{CaO} \cdot \text{SiO}_2$) + 2 types of GB.

Microstructural variables

◆ Density

- crystallographic (from unit cell parameters)
- theoretical (zero porosity, takes into account real composition)
- apparent (geometrical) density ($<$ theoretical)
- relative density = $(\text{apparent}/\text{theoretical}) \times 100$

◆ Porosity

- closed (only closed above 93% r.d.)
- open (pore networks connected to the surface)
- intragranular; - intergranular

◆ Grain size (simplest method: mean intercept length)

◆ Grain size distribution (monomodal, bimodal, abnormal grain growth)

◆ Grain shape (equiaxed, elongated, prismatic, columnar, tabular, platelets)

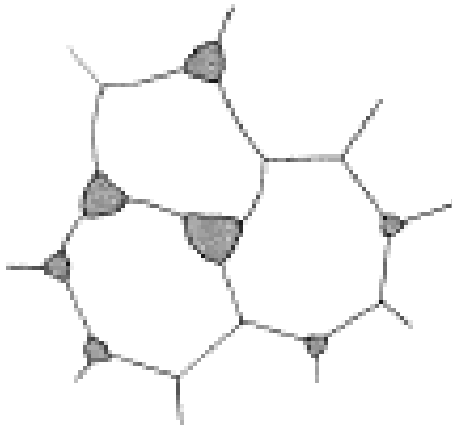
- aspect ratio (ratio longest/shortest size dimensions)

◆ Extended defects (dislocations, stacking faults, twins, domain walls)

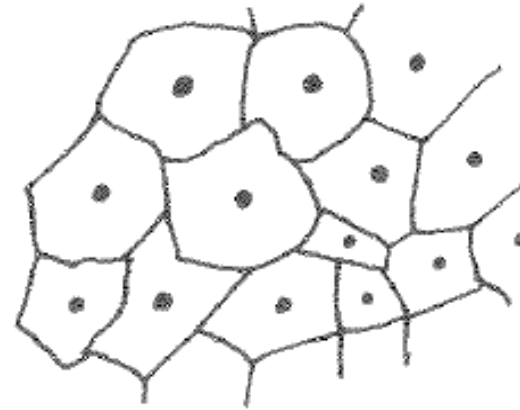
◆ Second phase composition, shape and distribution

◆ Texture (grains oriented in a preferential direction)

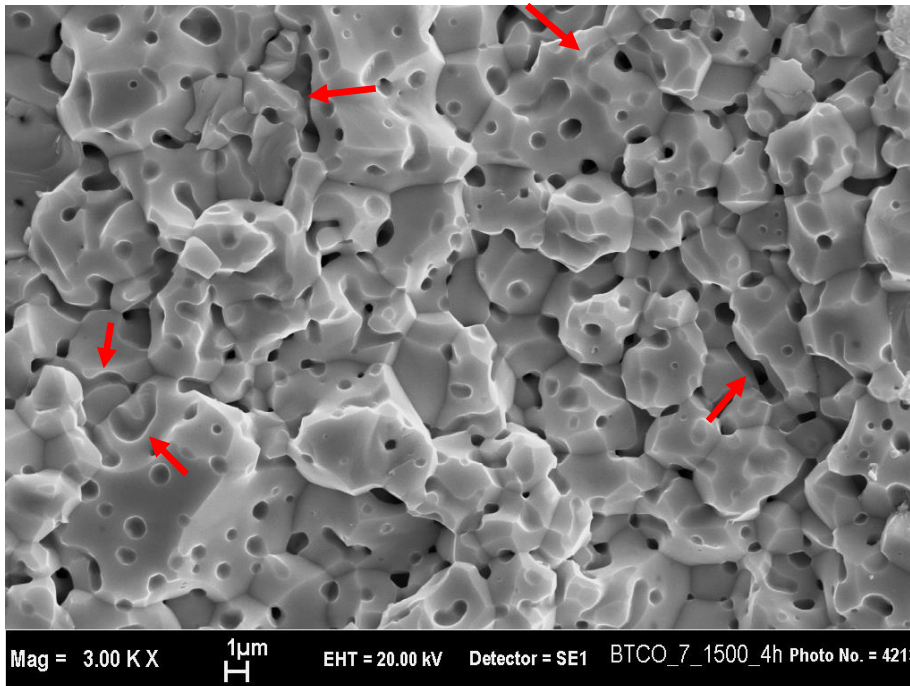
Types of Porosity



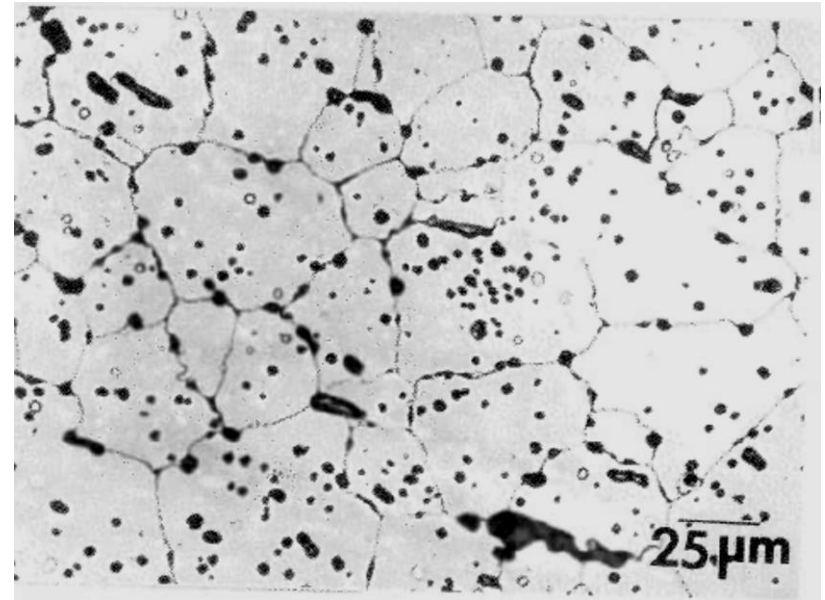
Intergranular porosity



Intragranular porosity



Ba(Ti,Ce)O₃ ceramic

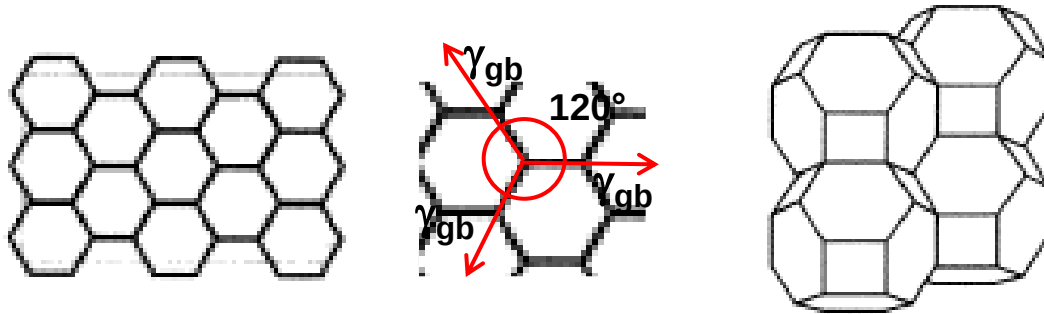


Y₂O₃:ZrO₂ (PSZ) ceramic

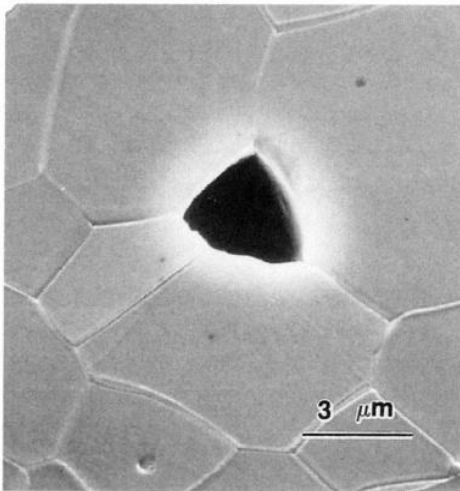
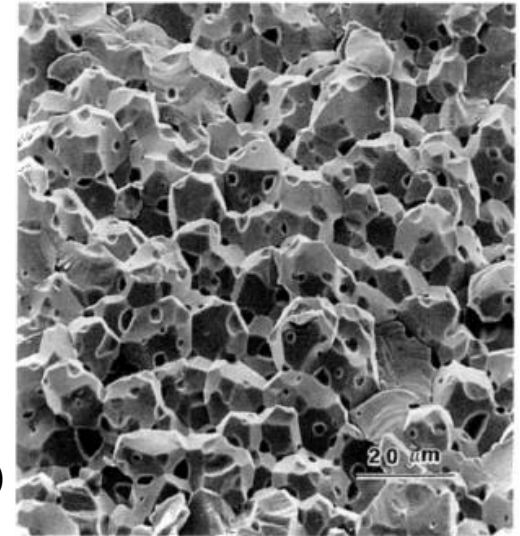
Intra- and intergranular porosity

Closed (intragranular) and open (intergranular channels) porosity

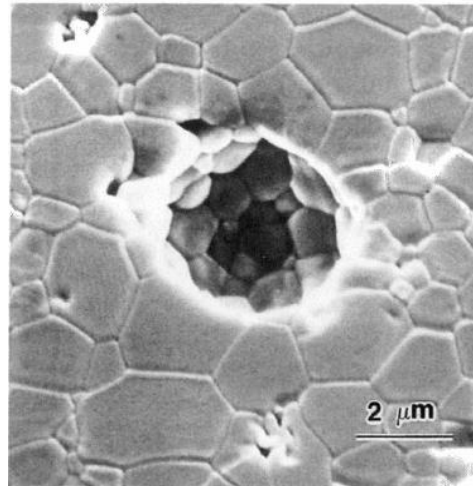
Shape of grains and pores



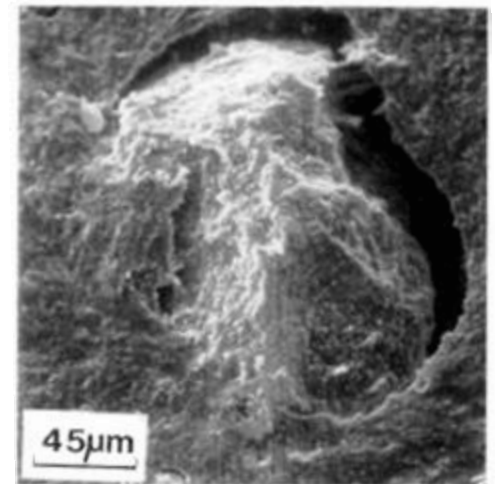
Equilibrium shape of grains. Hexagons (2D) and truncated octahedron (3D)



Concave pore



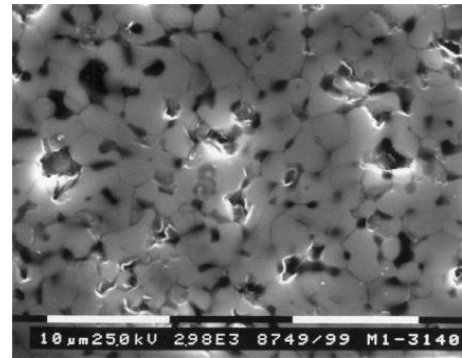
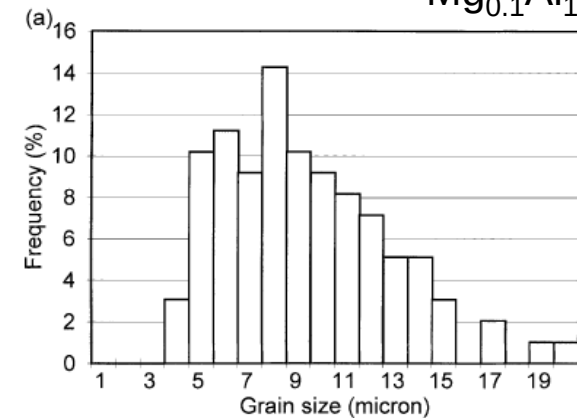
Convex pore



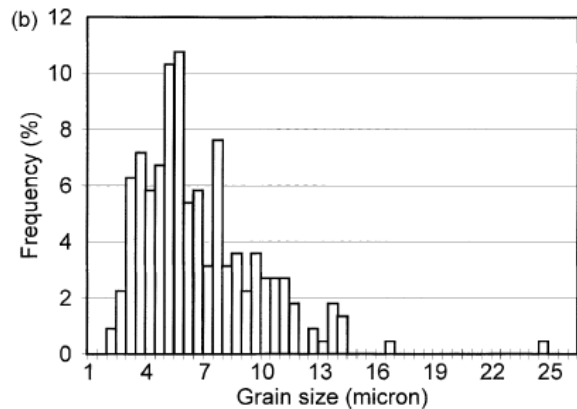
Irregular pore
associated with a
hard agglomerate

Grain size distributions and abnormal grain growth

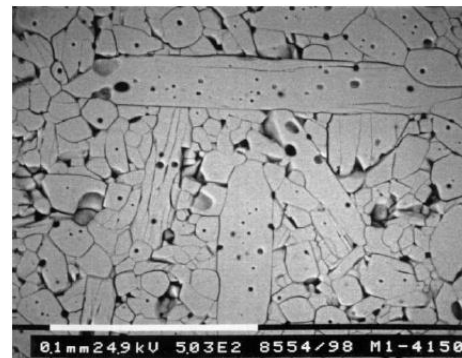
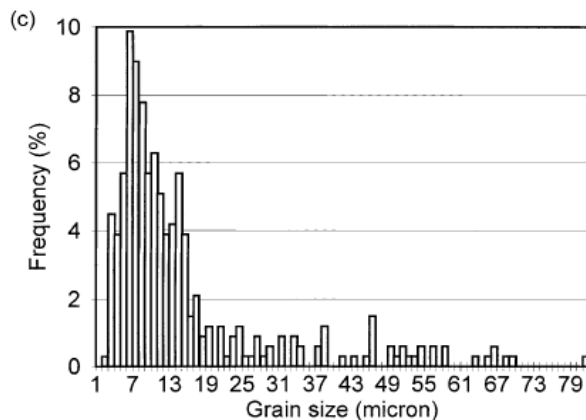
$\text{Mg}_{0.1}\text{Al}_{1.8}\text{Ti}_{1.1}\text{O}_5$ ceramics



1450°C/2 h
Equiaxed grains with
monomodal distribution



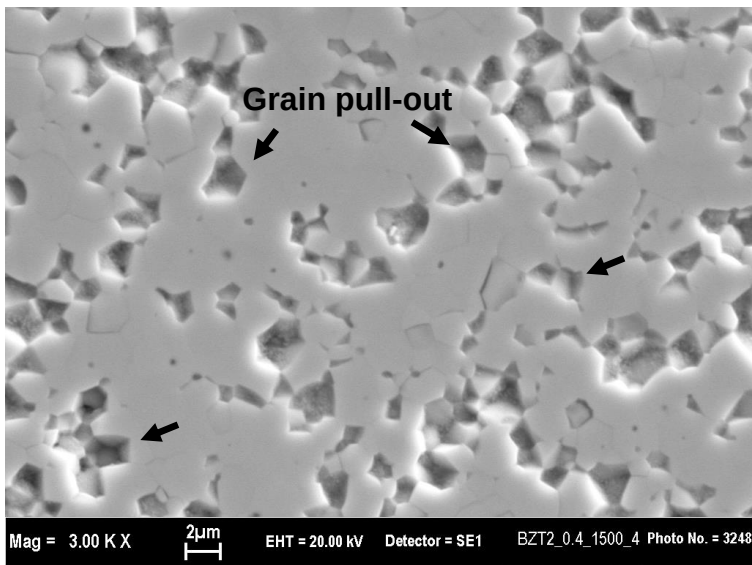
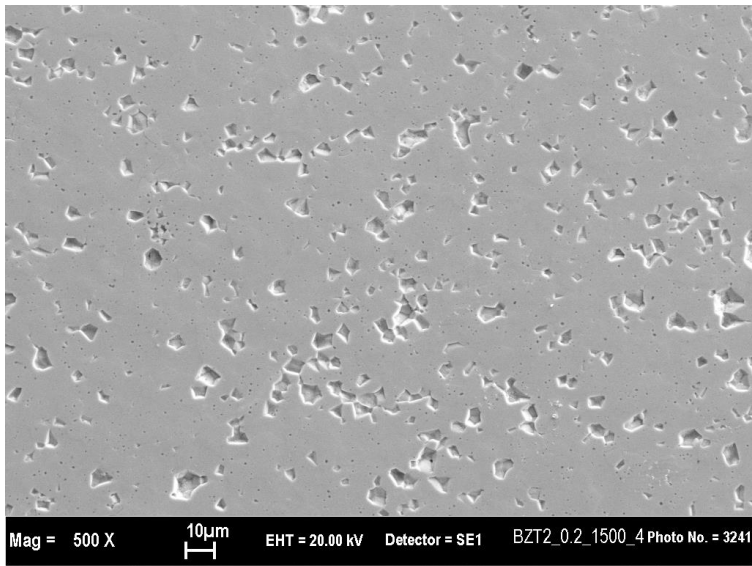
1500°C/2 h
Some large elongated grains
appear: onset of AGG



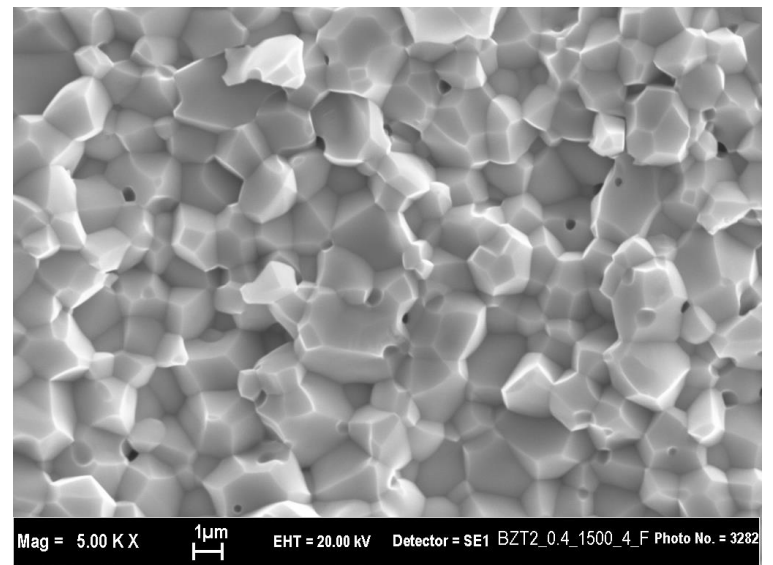
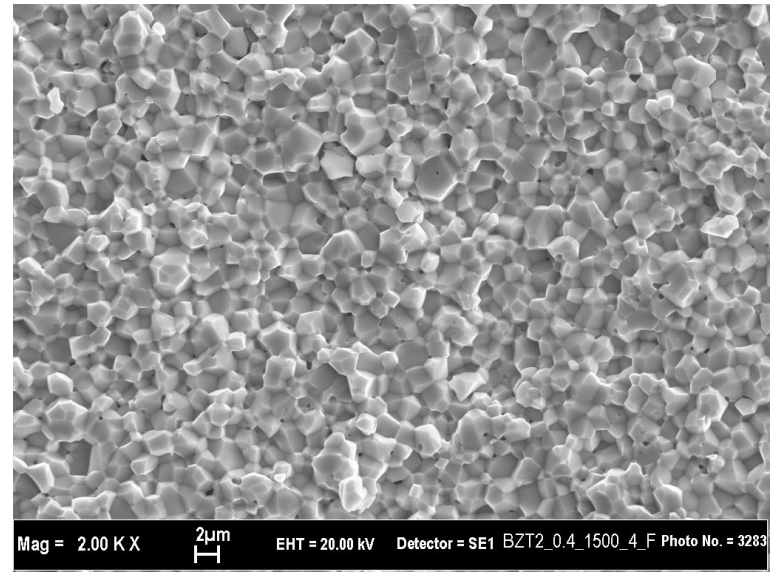
1550°C/2 h
Bimodal distribution related to
AGG

Revealing Microstructure

Polished surface



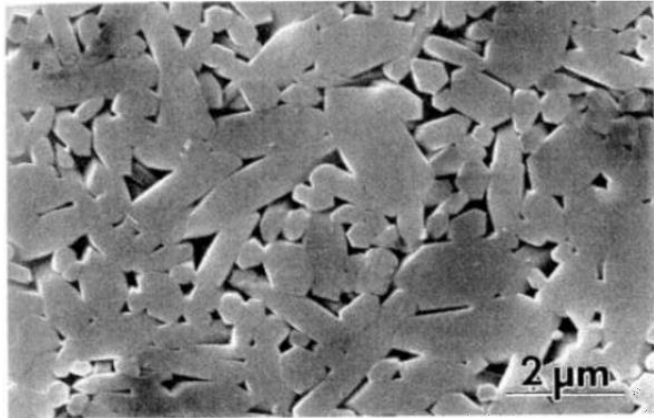
Fractography



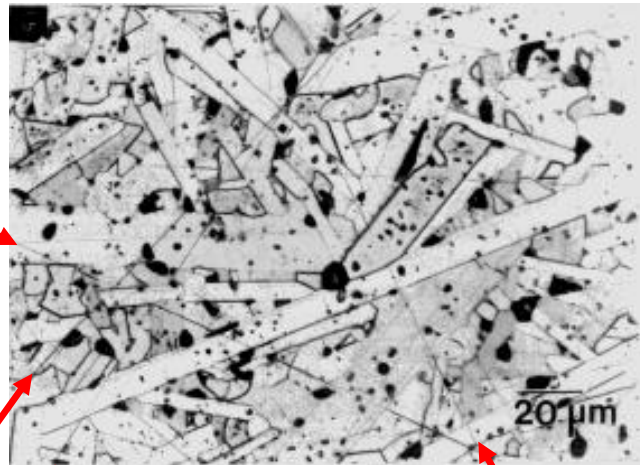
Cross-sections of Ba(Ti,Zr)O₃ ceramics

Revealing Microstructure

Chemical etching

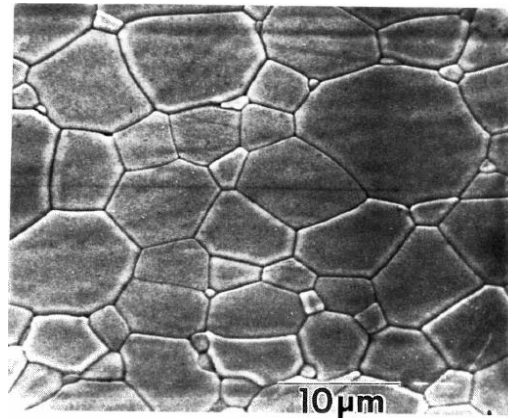


Si_3N_4

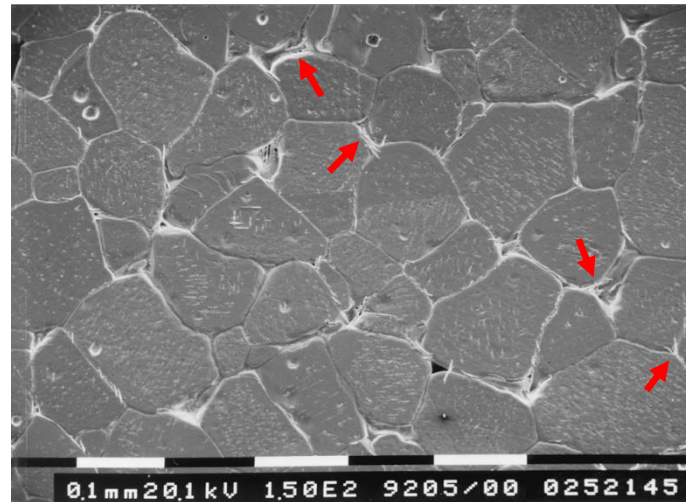


SiC

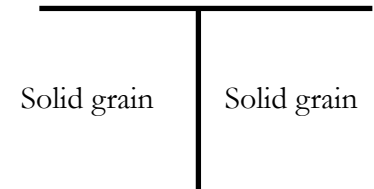
Thermal etching Vapour



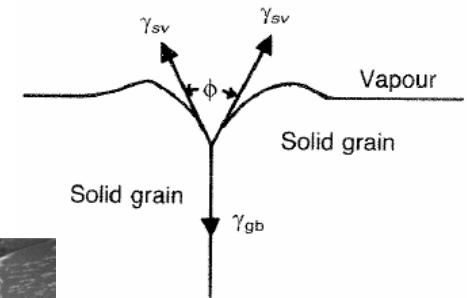
Fully-dense alumina



Er-doped BaTiO_3



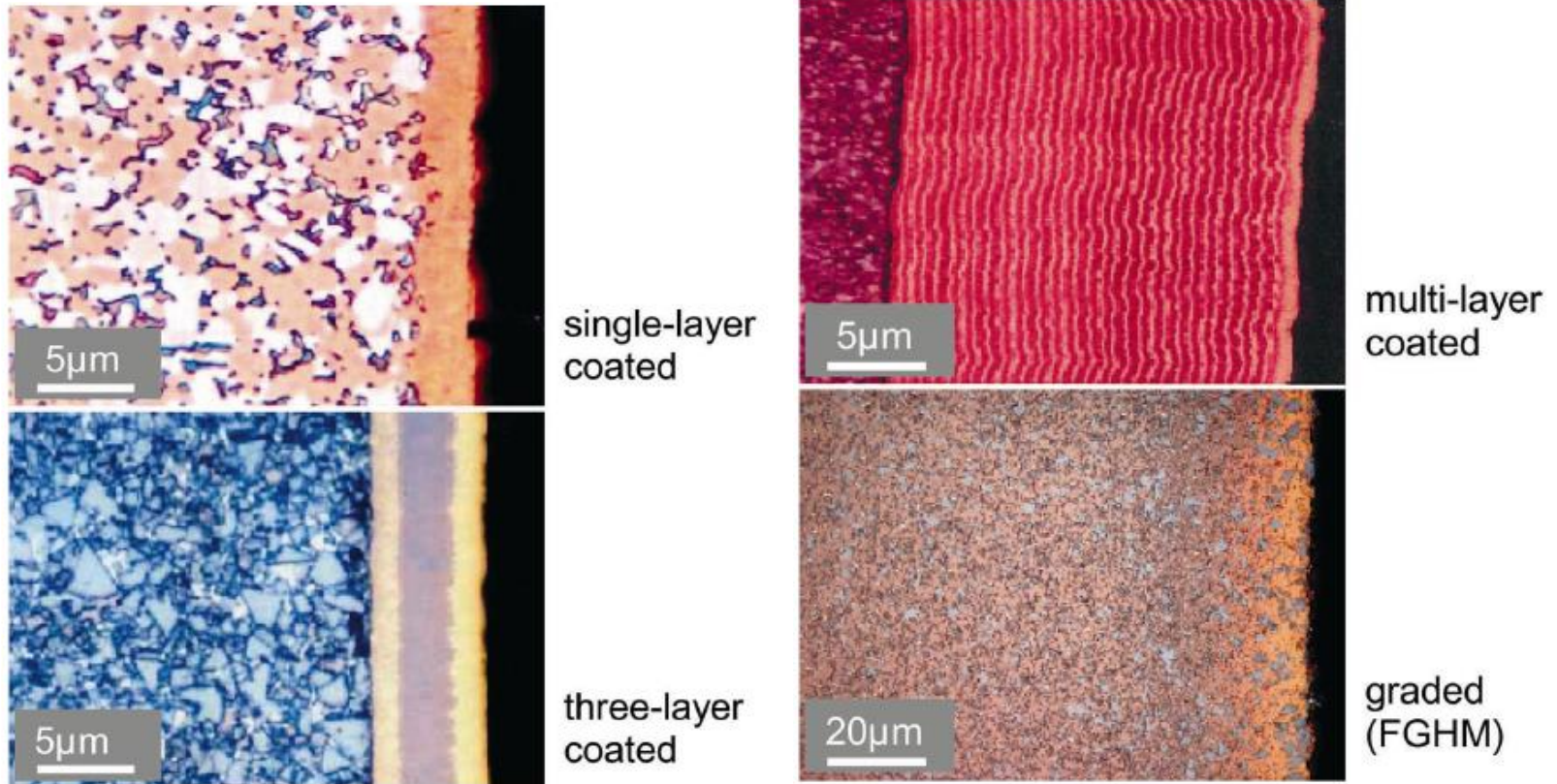
After polishing



After thermal treatment

Revealing Microstructure

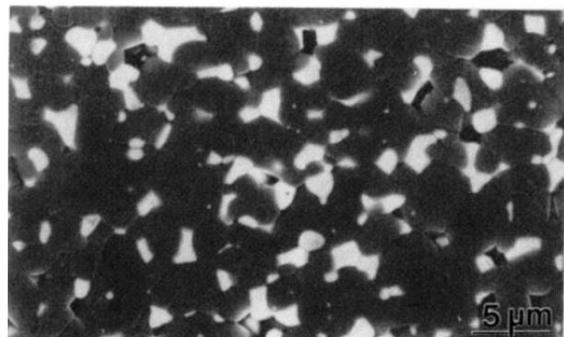
Optical microscopy in polarized light



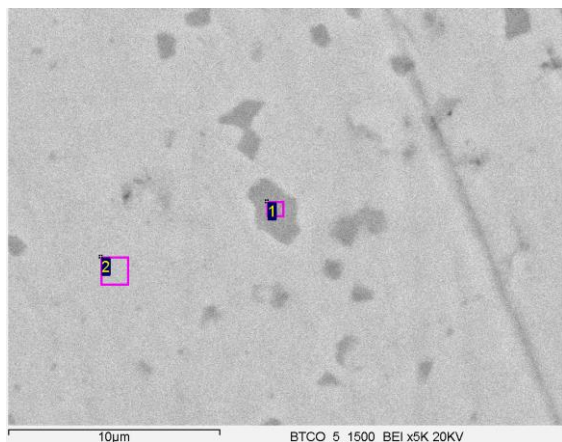
Coated Co:WC:Ti(C,N) cermets

Revealing Microstructure

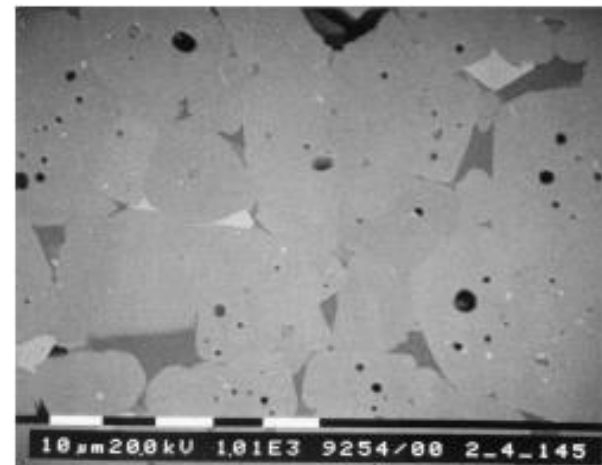
Scanning electron microscopy using backscattered electrons (BEI)



Al_2O_3 -(Zr,Y) O_2 composite
The darker phase is alumina



$\text{Ba}(\text{Ti,Ce})\text{O}_3$ ceramic
containing
inclusions
Ce-rich



Secondary phases in Er-doped BaTiO_3
White phase: $\text{Er}_2\text{Ti}_2\text{O}_7$
Dark gray phase: $\text{Ba}_6\text{Ti}_{17}\text{O}_{40}$

Spectrum	In stats.	O	Ti	Ba	Ce	Total
1	Yes	23.43	9.37	58.72	8.48	100.00
2	Yes	22.41	2.77	41.71	33.11	100.00